

PNSB BSC KPI — Database Design

Reflects the actually-built schema (verified against the live MySQL `information_schema`).

Stack: Laravel 13 + Livewire 4 (Volt) + MySQL. Spatie Permission for roles; Spatie Media Library for attachments; owen-it/laravel-auditing for the audit trail.

Last updated: 2026-06-30 (added the `audits` table + audit-trail feature)

Design Principles

- All scoring rules are **ALP Board ketetapan** — stored as data (templates, perspective weights, bell-curve targets), not hardcoded.
- `manager_id` chain on `users` drives the entire review hierarchy.
- **Snapshot architecture:** `scorecard_kpis` / `scorecard_competencies` copy the KPI/competency definition (title, uom, type...) at assignment time, so a historical scorecard stays stable even if the library is later edited. `kpi_id` / `competency_item_id` are kept (nullable) as a soft link back to the source.
- **Period-based scoring:** each scorecard KPI/competency is scored once per `review_period` (Mid-Year, Year-End), then aggregated into the line on `scorecard_kpis` / `scorecard_competencies`, then into the scorecard totals.
- Status progression is one-way and logged on every change (`scorecard_status_logs`).
- Enums for ALP-locked values; VARCHAR for anything HR might extend later.
- `softDeletes` on the master/reference tables (users, departments, designations, kpis, scorecards, cycles, perspectives, templates, competency_items, review_periods, strategic_objectives, bell_curve_targets).
- **General audit trail:** `owen-it/laravel-auditing` records create/update/delete on all domain & config models, plus authentication events (login/logout/failed-login), into the polymorphic `audits` table. The hand-rolled `moderation_logs` / `score_overrides` / `scorecard_status_logs` remain the **domain-specific** trails and are deliberately **not** double-audited.

Roles & Categories

Roles are managed by **Spatie Permission** (`roles` , `permissions` , `model_has_roles` , `model_has_permissions` , `role_has_permissions`) — there is **no** `role` **column** on

`users`. The 5 system roles:

Role	Purpose	Appraised?
<code>super-admin</code>	Full system admin; stands in for HR at moderation/lock	No
<code>hr-admin</code>	HR operations, moderation MOD1, lock	No
<code>executive-director</code>	CEO / KPE — top of chain, runs moderation MOD2	Yes (CEO template)
<code>division-head</code>	Ketua Bahagian — reviews subordinates and is reviewed	Yes (Executive template)
<code>staff</code>	Executive or Support employees	Yes

`users.category` (`ceo` / `executive` / `support`) is separate from role — it selects the scorecard template / scoring formula. A `division-head` is `category = executive`. Admin roles (`super-admin` , `hr-admin`) are excluded from scorecard generation regardless of category.

Tables

1. `departments`

```
id          bigint PK
name        varchar(150)
code        varchar(50) UNIQUE
parent_id   bigint NULL → departments.id
head_user_id bigint NULL → users.id (set after users created)
is_active   boolean
deleted_at, timestamps
```

2. `designations` (new since planning — replaces free-text job_title)

```
id          bigint PK
name        varchar(150)
short_name   varchar(50)          -- e.g. CEO, FM, FE, ITM, HRM
department_id bigint → departments.id
```

```
description    text NULL
is_active      boolean
deleted_at, timestamps
```

3. `users`

```
id            bigint PK
name          varchar(255)
employee_no   varchar(50) UNIQUE
email         varchar(255) UNIQUE
department_id bigint NULL → departments.id
designation_id bigint NULL → designations.id
manager_id    bigint NULL → users.id    (self-referential; NULL = top of chain / admin)
category      enum('ceo','executive','support') -- drives scoring template
is_active     boolean
email_verified_at timestamp NULL
password      varchar(255)
remember_token varchar(100) NULL
deleted_at, timestamps
```

Permissions come from Spatie role assignments, **not** a column here.

4. `appraisal_cycles`

```
id          bigint PK
year        year UNIQUE
status      enum('draft','active','appraisal','moderation','closed') -- one-way lifecycle
opened_by   bigint → users.id
opened_at   timestamp NULL
closed_at   timestamp NULL
deleted_at, timestamps
```

5. `review_periods` (new — half-yearly scoring windows under a cycle)

```
id          bigint PK
cycle_id    bigint → appraisal_cycles.id
sequence    tinyint          -- 1, 2
label       varchar(255)      -- 'Mid-Year', 'Year-End'
weight      decimal(5,2)      -- 50.00 each
is_final    boolean           -- the period that closes the year
```

```

opens_at    timestamp NULL
closes_at   timestamp NULL
status      enum('upcoming','open','scored','closed')
deleted_at, timestamps

```

6. `bsc_perspectives`

```

id          bigint PK
name        varchar(100)          -- Financial / Customer / Internal Business Process /
description text NULL
weight      decimal(5,2)          -- 40 / 30 / 20 / 10
sort_order  tinyint
deleted_at, timestamps

```

Seed (ALP-locked): Financial 40 · Customer 30 · Internal Business Process 20 · Learning & Growth 10.

7. `strategic_objectives` (*new — objectives under each perspective*)

```

id          bigint PK
bsc_perspective_id bigint → bsc_perspectives.id
name        varchar(150)
description  text NULL
sort_order  tinyint
is_active   boolean
deleted_at, timestamps

```

8. `kpis`

```

id          bigint PK
title       varchar(255)
bsc_perspective_id bigint → bsc_perspectives.id
strategic_objective_id bigint NULL → strategic_objectives.id
uom         varchar(100)          -- RM, %, Hari, Bil, Tarikh...
pengukuran  varchar(255) NULL    -- measurement description
weight      decimal(5,2)          -- suggested default weight
level       enum('company','department','individual') -- cascade scope
is_active   boolean

```

```

created_by          bigint → users.id
deleted_at, timestamps

```

Cascade scoping pivots:

- `kpi_department` (`kpi_id, department_id`) — for `level = department`
- `kpi_designation` (`kpi_id, designation_id`) — for `level = individual`
- `level = company` cascades to every KPI-based employee (no pivot rows).

9. `scorecard_templates`

```

id                bigint PK
category          enum('ceo','executive','support') UNIQUE
kpi_weight        decimal(5,2)
competency_weight decimal(5,2)
competency_method varchar(50) NULL  -- currently 'average'
deleted_at, timestamps

```

Seed (ALP-locked): ceo 100/0 · executive 80/20 · support 0/100.

10. `competency_items`

```

id                bigint PK
title             varchar(255)
description        text NULL
type              enum('core','functional')
level_descriptions json NULL      -- Likert level text (1..n descriptions)
is_active          boolean
deleted_at, timestamps

```

Scoping pivots: `competency_item_department`, `competency_item_designation` (functional competencies scoped to dept/designation; core apply to all).

11. `scorecards`

```

id                bigint PK
user_id           bigint → users.id
cycle_id          bigint → appraisal_cycles.id
template_id       bigint → scorecard_templates.id
status            enum('draft','kpi_submitted','kpi_approved','appraisal','moderat
kpi_score         decimal(5,2) NULL  -- Σ weighted_score of scorecard_kpis

```

```

competency_score      decimal(5,2) NULL      -- AVG final_rating of scorecard_competenc
competency_self_comment text NULL          -- employee overall self-assessment commen
final_score           decimal(5,2) NULL      -- (kpi_score*kpi_weight + competency_sco
grade                 enum('excellent','very_good','good','satisfactory','needs_improv
locked_at             timestamp NULL
locked_by             bigint NULL → users.id
deleted_at, timestamps
UNIQUE (user_id, cycle_id)

```

Competency attachments: stored via Spatie Media Library (`media` table, collection `competency_attachments`).

Grade auto-assignment (C5, on `final_score`): ≥90 excellent · ≥76 very_good · ≥60 good · ≥50 satisfactory · <50 needs_improvement.

12. `scorecard_kpis` (*snapshot of an assigned KPI*)

```

id                bigint PK
scorecard_id      bigint → scorecards.id
kpi_id            bigint NULL → kpis.id      -- soft link to source
bsc_perspective_id bigint → bsc_perspectives.id (denormalised)
kpi_title         varchar(255)              -- snapshot
objective_name    varchar(255) NULL         -- snapshot
uom              varchar(100)              -- snapshot
pengukuran       varchar(255) NULL         -- snapshot
kpi_weight        decimal(5,2)              -- Jumlah Kecil Pemberat
threshold        varchar(100) NULL         -- Ambangan
meet_target      varchar(100) NULL         -- Setuju
stretched        varchar(100) NULL         -- Lebihan
actual          varchar(100) NULL         -- aggregated raw value
tier_score       decimal(5,2) NULL         -- 0 / 60 / 80 / 100
weighted_score   decimal(5,2) NULL         -- tier_score/100 × kpi_weight
notes           text NULL
timestamps

```

Tier scoring (C7): Below Threshold 0 · Threshold 60 · Meet Target 80 · Stretched 100.

13. `scorecard_kpi_period_scores` (*new — per review-period KPI actuals*)

```

id                bigint PK
scorecard_kpi_id  bigint → scorecard_kpis.id

```

```

review_period_id  bigint → review_periods.id
actual            varchar(100) NULL
tier_score        decimal(5,2) NULL
weighted_score    decimal(5,2) NULL
notes            text NULL
timestamps

```

14. `scorecard_competencies` *(snapshot of an assessed competency)*

```

id                bigint PK
scorecard_id      bigint → scorecards.id
competency_item_id  bigint NULL → competency_items.id  -- soft link to source
competency_title   varchar(255)                        -- snapshot
competency_type    enum('core','functional')            -- snapshot
self_rating        decimal(5,2) NULL                    -- Penilaian Kendiri
manager_rating     decimal(5,2) NULL                    -- Penilaian Pengurus (override)
final_rating       decimal(5,2) NULL
manager_justification text NULL
timestamps

```

15. `scorecard_competency_period_scores` *(new — per review-period competency ratings)*

```

id                bigint PK
scorecard_competency_id  bigint → scorecard_competencies.id
review_period_id  bigint → review_periods.id
self_rating        decimal(5,2) NULL
manager_rating     decimal(5,2) NULL
final_rating       decimal(5,2) NULL
manager_justification text NULL
timestamps

```

16. `bell_curve_targets`

```

id                bigint PK
cycle_id          bigint → appraisal_cycles.id
grade            enum('excellent','very_good','good','satisfactory','needs_improvement')
target_count     int
target_pct       decimal(5,2)

```

```
timestamps, deleted_at
UNIQUE (cycle_id, grade)
```

Targets scale to headcount (largest-remainder). Reference distribution (per ~91 staff): excellent 3.3% · very_good 17.58% · good 67.03% · satisfactory 7.69% · needs_improvement 4.4%.

17. `moderation_logs`

```
id          bigint PK
scorecard_id bigint → scorecards.id
cycle_id    bigint → appraisal_cycles.id
round       enum('MOD1','MOD2')      -- MOD1 = HR per-dept, MOD2 = KPE org-wide
before_score / after_score decimal(5,2) NULL
before_grade / after_grade enum(...grades...) NULL
moderated_by bigint → users.id
notes       text NULL
moderated_at timestamp
timestamps
```

A moderation "adjust" moves the grade band; the raw score is preserved, every move logged here.

18. `score_overrides` (audit of manager/verification overrides)

```
id          bigint PK
scorecard_id bigint → scorecards.id
override_type enum('kpi','competency')
stage       varchar(30)      -- e.g. verification / appraisal
field       varchar(50) NULL
field_ref_id bigint NULL     -- scorecard_kpis.id or scorecard_competencies.id
before_value varchar(255) NULL
after_value  varchar(255)
overridden_by bigint → users.id
notes       text NULL
overridden_at timestamp
timestamps
```

19. `scorecard_status_logs` (full status-change audit; feeds dashboard "Recent Activity")

```

id          bigint PK
scorecard_id bigint → scorecards.id
from_status varchar(50) NULL    -- NULL = initial creation
to_status   varchar(50)
changed_by  bigint → users.id
notes       text NULL
changed_at  timestamp
timestamps

```

20. `media` (*Spatie Media Library — competency evidence files*)

Standard Spatie schema (`model_type`, `model_id`, `collection_name`, `file_name`, `mime_type`, `disk`, `size`, json props...). Collection `competency_attachments` is attached to `Scorecard`.

21. `audits` (*general audit trail — owen-it/laravel-auditing*)

```

id          bigint PK
user_type    varchar NULL      -- actor morph (App\Models\User); NULL for system
user_id      bigint NULL
event        varchar           -- created/updated/deleted/restored + login/logout
auditable_type varchar NULL    -- audited model morph; NULL for model-less auth
auditable_id bigint NULL
old_values   json NULL         -- changed attributes before
new_values   json NULL         -- changed attributes after (secrets excluded: p
url          varchar NULL      -- request URL, or 'console'
ip_address   varchar(45) NULL
user_agent   varchar(1023) NULL
tags         varchar NULL      -- 'auth' for login/logout/login_failed
timestamps
INDEX (auditable_type, auditable_id)

```

Audited models (16): appraisal_cycles, bell_curve_targets, bsc_perspectives, competency_items, departments, designations, kpis, review_periods, scorecards, scorecard_competencies, scorecard_competency_period_scores, scorecard_kpis, scorecard_kpi_period_scores, scorecard_templates, strategic_objectives, users.

NOT audited (already append-only logs — would be recursive): `moderation_logs`, `score_overrides`, `scorecard_status_logs`.

Auth events: login/logout attach to the user (so they appear in that user's history); failed logins are **system-level** rows (null `auditable`) carrying the attempted email + IP. Auditing is **OFF** in

console (seeders/tinker don't record — controlled by `audit.console`). Viewer at `/audit-trail` is **super-admin only** (separation of duties: the log can record HR's own actions).

Framework / package tables

`roles`, `permissions`, `model_has_roles`, `model_has_permissions`, `role_has_permissions` (Spatie Permission); plus standard Laravel `migrations`, `sessions`, `cache`, `jobs`, `password_reset_tokens`.

Relationships Summary

```
departments —< designations
departments —< users (department_id)
designations —< users (designation_id)
users —< users (manager_id - self-referential hierarchy)

appraisal_cycles —< review_periods
appraisal_cycles —< scorecards
appraisal_cycles —< bell_curve_targets
appraisal_cycles —< moderation_logs

bsc_perspectives —< strategic_objectives —< kpis
bsc_perspectives —< kpis
kpis >—< departments (kpi_department)
kpis >—< designations (kpi_designation)

users —< scorecards >— scorecard_templates
scorecards —< scorecard_kpis —< scorecard_kpi_period_scores >— review_periods
scorecards —< scorecard_competencies —< scorecard_competency_period_scores >— review_periods
kpis —< scorecard_kpis (soft link, nullable)
bsc_perspectives —< scorecard_kpis (denormalised)
competency_items —< scorecard_competencies (soft link, nullable)
competency_items >—< departments / designations (scoping pivots)

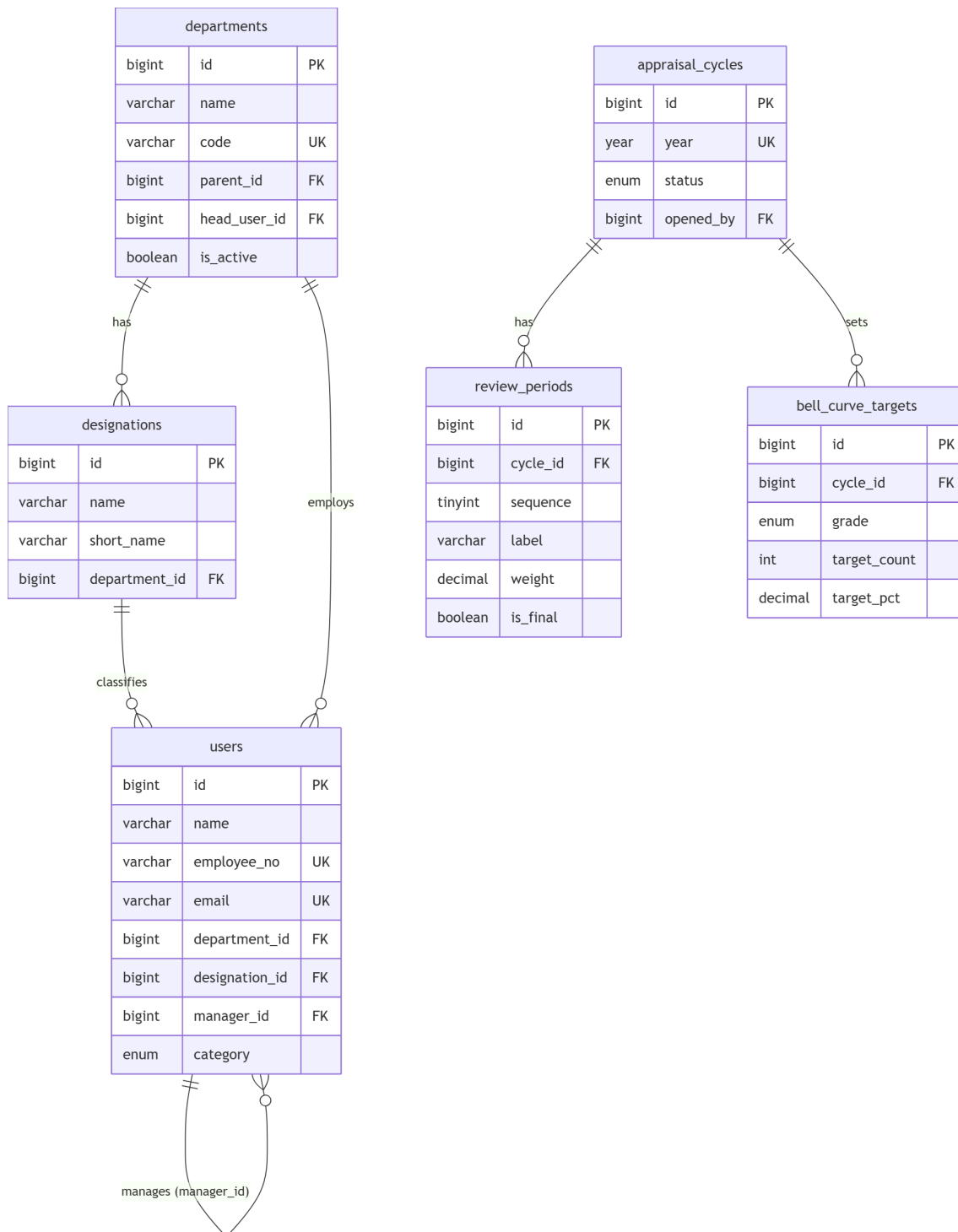
scorecards —< moderation_logs / score_overrides / scorecard_status_logs
scorecards —< media (competency_attachments)

audits >— (polymorphic) auditable: any of the 16 audited models (NULL for failed-logins)
audits >— (polymorphic) user: users (the actor; NULL = system)
```

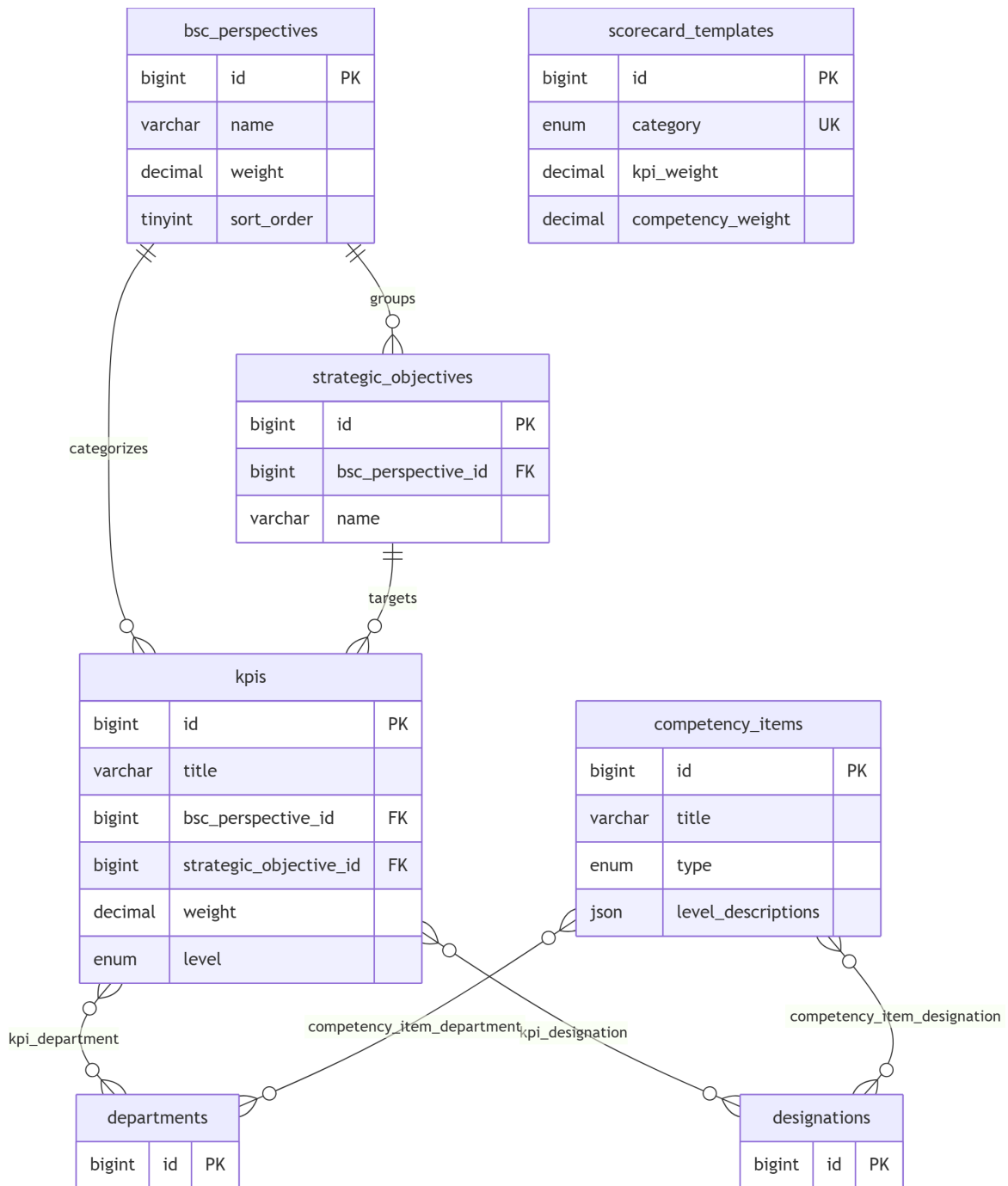
Entity-Relationship Diagrams

Split into three focused views so each renders large and readable (one 22-table graph shrinks to nothing on a page). Pivots are drawn as many-to-many links; entities shown as `id`-only boxes are anchors whose full definition lives in another view.

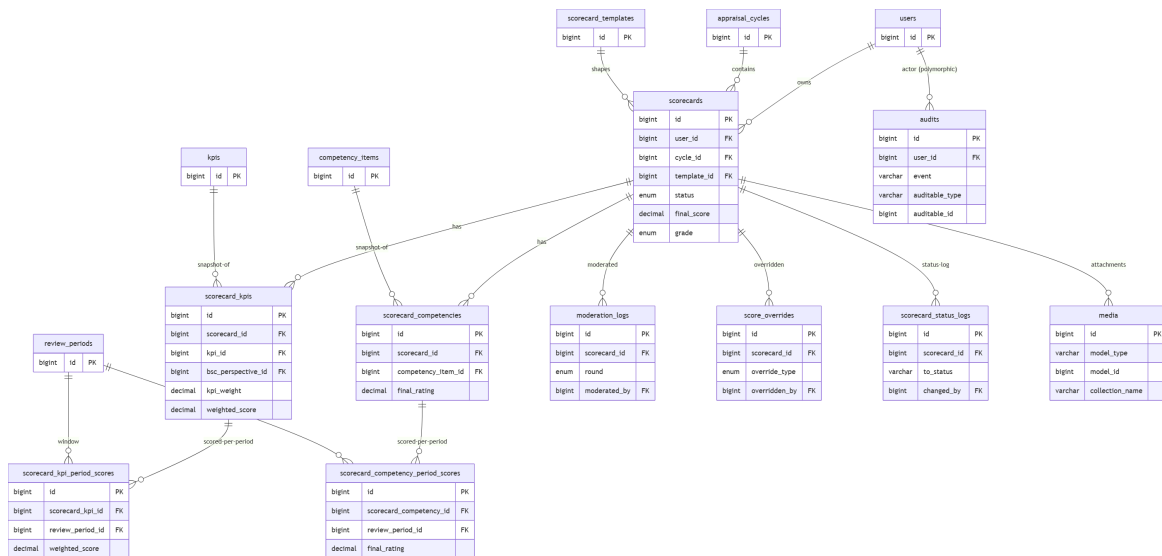
ERD 1 — People, Org & Cycles



ERD 2 — KPI & Competency Library



ERD 3 — Scorecards & Scoring



Score Flow (how a number becomes a grade)

- Per review period** → `scorecard_kpi_period_scores.weighted_score` & `scorecard_competency_period_scores.final_rating` captured.
- Aggregate to line** → `scorecard_kpis.weighted_score` (period-weighted) & `scorecard_competencies.final_rating`.
- Aggregate to scorecard** → `kpi_score` = $\sum \text{weighted_score}$; `competency_score` = AVG `final_rating`.
- Final** → `final_score` = $(\text{kpi_score} \times \text{kpi_weight} + \text{competency_score} \times \text{competency_weight}) / 100$ (all 0–100 scale).
- Grade** auto-assigned from C5 thresholds → optionally re-banded in moderation (score preserved).

Business Rules (application layer)

Rule	Where
<code>support</code> → no <code>scorecard_kpis</code> (0% KPI weight)	Scorecard generation
<code>ceo</code> → no <code>scorecard_competencies</code> (0% competency weight)	Scorecard generation
Admin roles (<code>super-admin</code> , <code>hr-admin</code>) excluded from scorecard generation	<code>eligibleEmployees()</code>
<code>manager_rating</code> becomes <code>final_rating</code> (C1)	Competency scoring

Rule	Where
<code>tier_score</code> only ever 0 / 60 / 80 / 100 (C7)	KPI scoring
<code>final_score</code> on 0–100 scale (not /5)	Score calc
Grade from <code>final_score</code> thresholds (C5)	Grade assignment
BSC weights sum to 100; KPI weights per perspective sum to that perspective's weight	Validation
Moderation MOD1 = <code>hr-admin</code> / <code>super-admin</code> ; MOD2 = <code>executive-director</code> / <code>super-admin</code>	Moderation gates
Scorecard status is one-way	Status transition guard

Notes vs the planning draft

- **Roles moved to Spatie** — the old `users.role` enum is gone.
- **Enums anglicised** — categories `ceo/executive/support`, grades `excellent...`
`needs_improvement` (were Malay).
- **Snapshot + period-score model added** — `scorecard_kpis` / `scorecard_competencies` now carry snapshot columns; new `*_period_scores` tables hold per-period entries; new `review_periods`.
- **Designations & strategic_objectives** added; `kpis` uses `title` / `weight` / `pengukuran` (no `title_my` / `category_scope`).
- `score_overrides` gained `stage` / `field`, values widened to varchar; `moderation_logs` carries `updated_at`.
- **Spatie Media Library** added for competency attachments.
- **Audit trail added (2026-06-30)** — `owen-it/laravel-auditing` `audits` table; 16 domain/config models audited + login/logout/failed-login; super-admin `/audit-trail` viewer. The three hand-rolled logs (`moderation_logs`, `score_overrides`, `scorecard_status_logs`) stay as-is and are not double-audited.